Deploying AI in Construction

Applications, Pilots & the Operating Model That Makes Them Work

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March 2026

Agenda

- The Challenge We Face
- Application Showcase
- The Vera Pilot
- Measuring Success
- Key Takeaways
- Q/A

The Challenge We Face

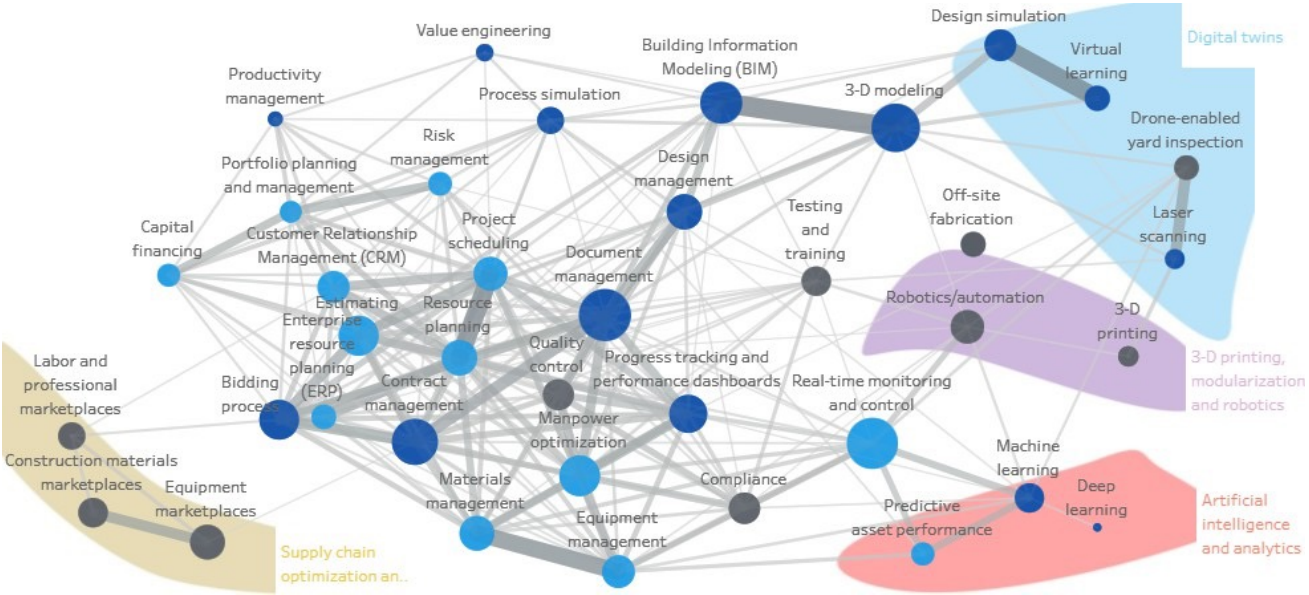
Construction workflows are complex, fragmented, and high-stakes

Image from esbnyc.com

Building Takes Longer Than It Used To

- The Empire State Building was built in **just over a year**
- In the intervening 100 years – there have been countless improvements.

  But those improvements have also increased project management and execution burden.



From McKinsey & Company

AI can be the bridge — *but it is easier said than done.*

Why AI Fails

> *We didn't start with 'we need to build an AI solution.' We started with 'here is the problem for persona X — how can we solve it?'*

Common Failure Patterns

- Ambiguous use cases
- No baseline comparison
- Feature sprawl
- No feedback loop

What Works

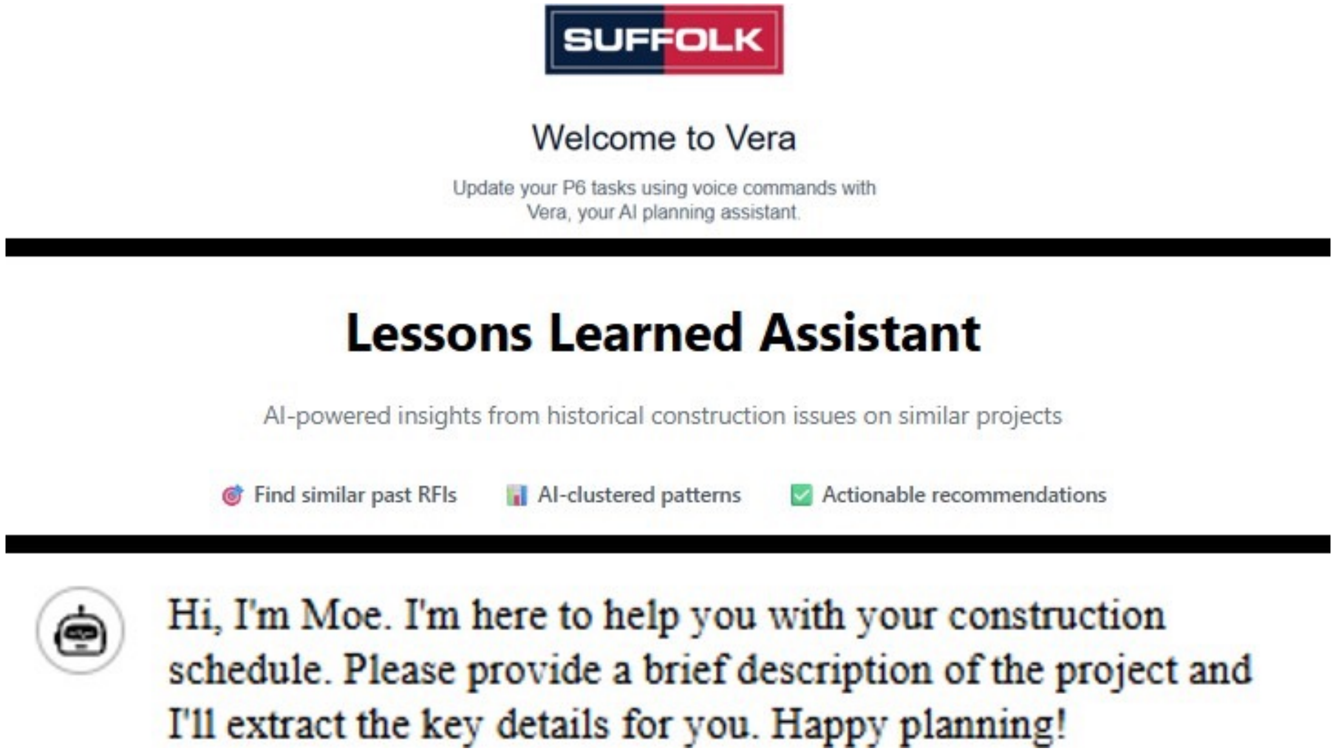
- Constrained pilots
- Clear personas
- Workflow-first design
- Measurable outcomes

Application Showcase

Real tools solving real workflow problems

Three Patterns of AI in Construction

1. **Streamliner**
2. **Knowledge Retrieval**
3. **Output Generator**



Vera is a streamliner, Lessons Learned is a Knowledge Retrieval tool, and Gen Schedule is an Output Generator

Vera

The Workflow

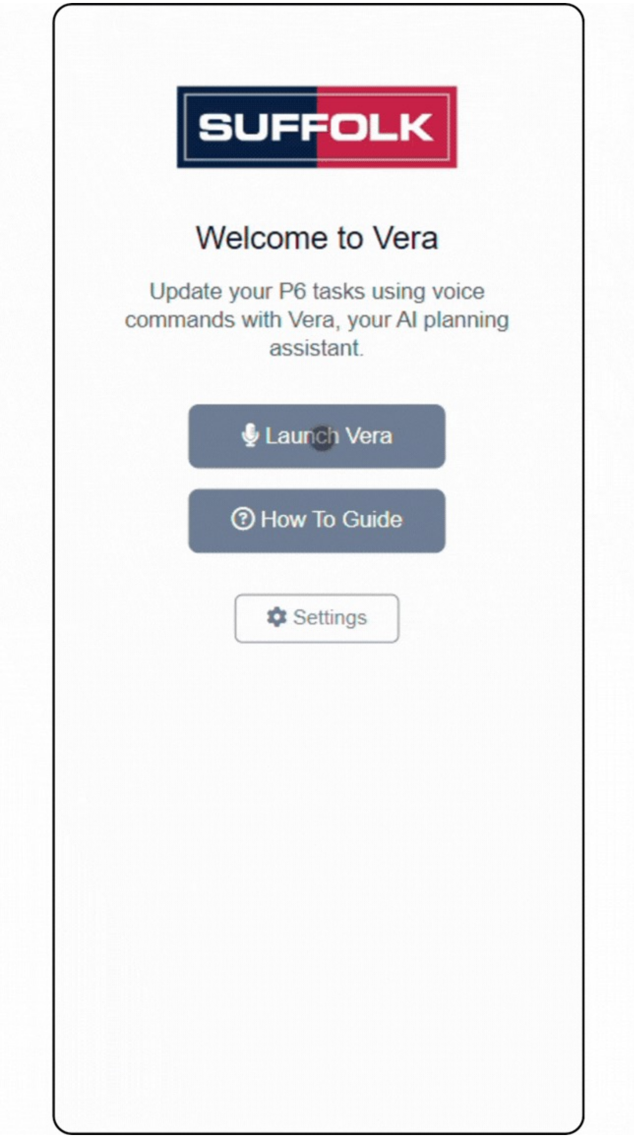
Schedule updates from the field

The Persona

Superintendents

What It Streamlines

Schedule Update



Vera streamlines field schedule updates through natural language

Lessons Learned

The Workflow

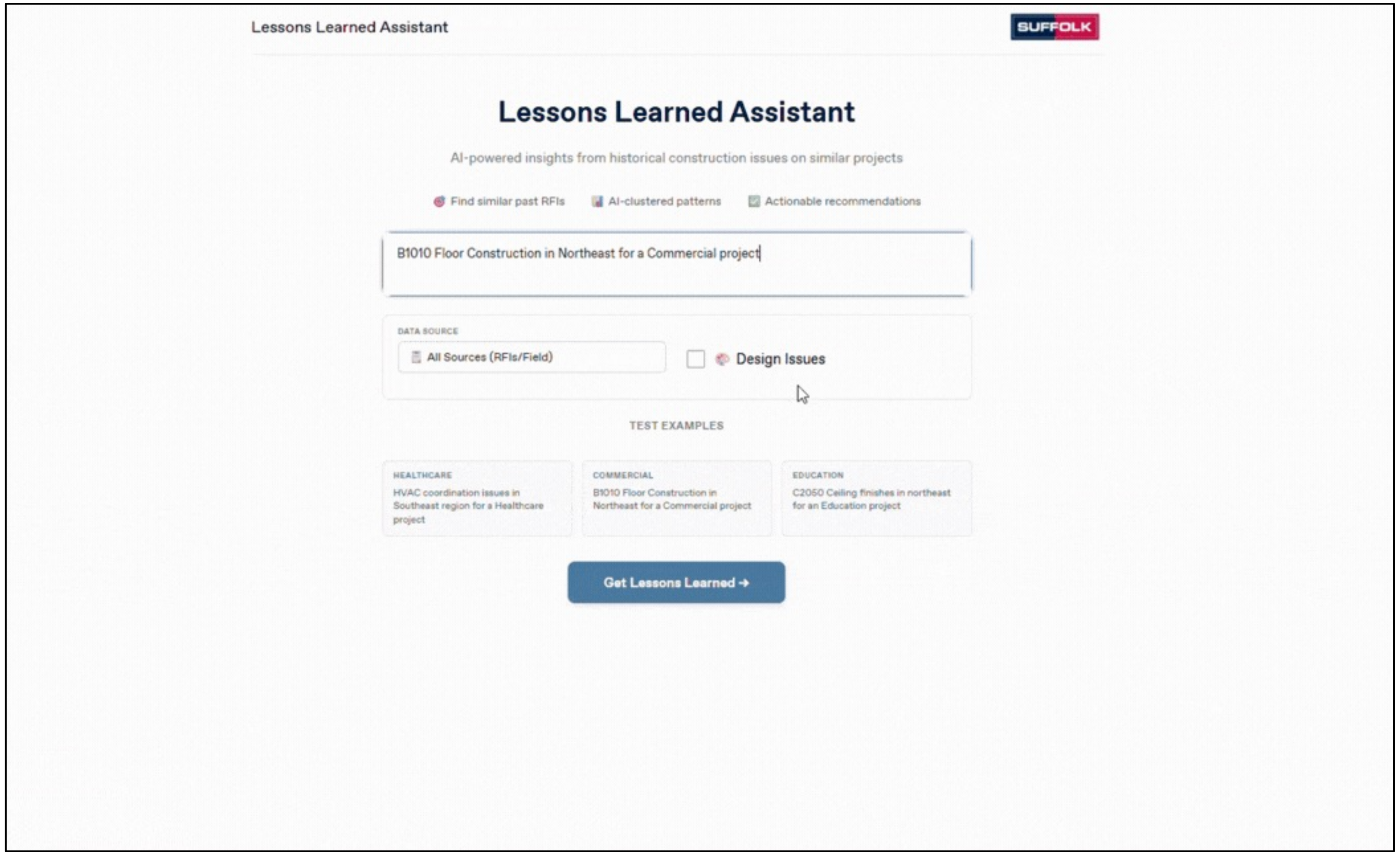
Manually capture lessons

The Persona

Construction teams

What It Streamlines

Knowledge retrieval



Lessons Learned clusters historical project lessons and allows you to query them to avoid the same mistakes

Generative Schedule

The Workflow

Initial Schedule Creation

The Persona

Planners

What It Streamlines

Schedule Generation

Gen Schedule lets you leverage LLMs to create an initial full schedule

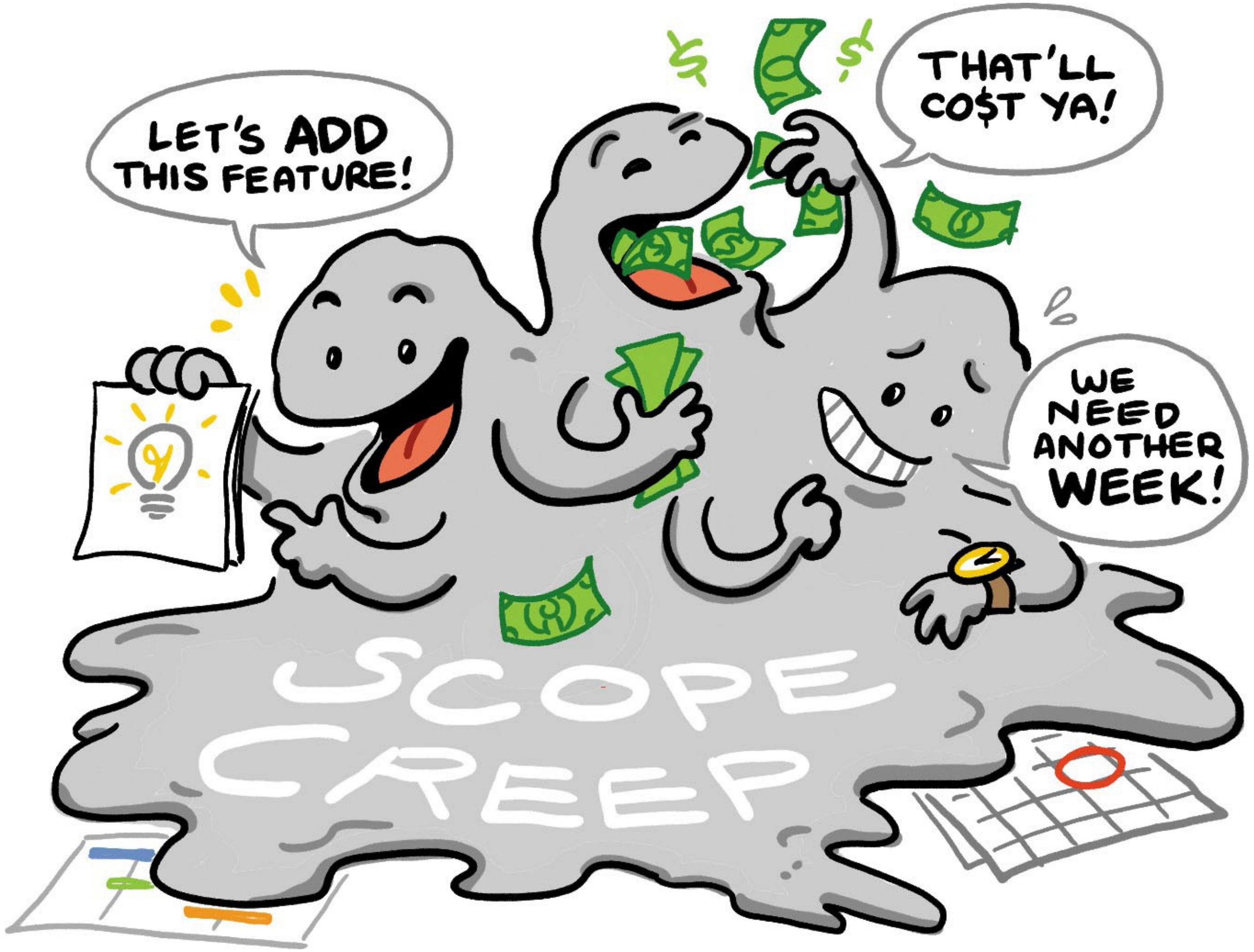
The Vera Pilot

Designing for success in a non-deterministic world

Pilot Success Essentials

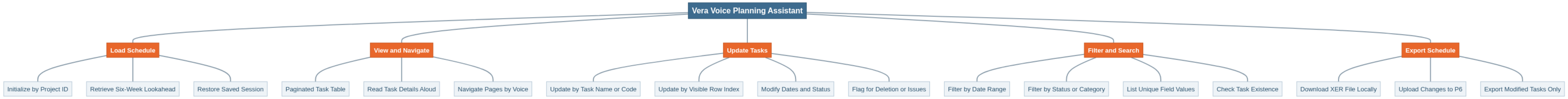
Design Principles

1. Clear Use Case
2. No feature sprawl
3. Defined persona



A focused pilot is not a limited one — it is a disciplined one - source: Rosenfeld Media

What is Vera?


Vera Voice Planning Assistant
Load Schedule
View and Navigate
Update Tasks
Filter and Search
Export Schedule
Initialize by Project ID
Retrieve Six-Week Lookahead
Restore Saved Session
Paginated Task Table
Read Task Details Aloud
Navigate Pages by Voice
Update by Task Name or Code
Update by Visible Row Index
Modify Dates and Status
Flag for Deletion or Issues
Filter by Date Range
Filter by Status or Category
List Unique Field Values
Check Task Existence
Download XER File Locally
Upload Changes to P6
Export Modified Tasks Only


Filter and Search
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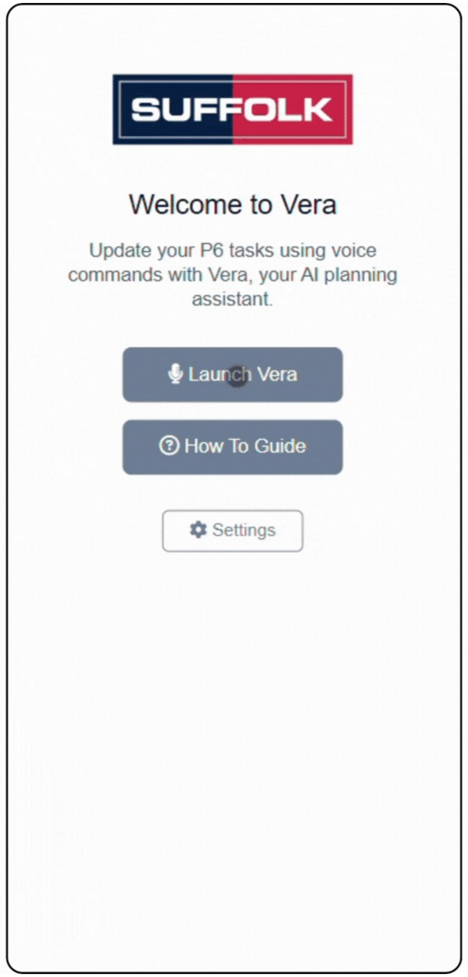
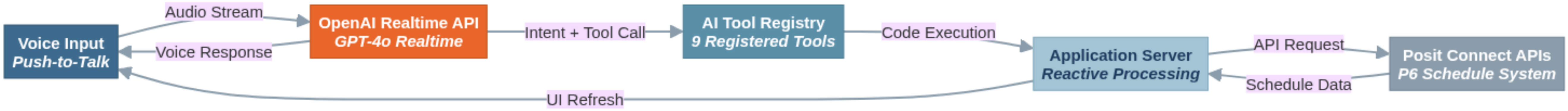


SUFFOLK
Welcome to Vera
Update your P6 tasks using voice commands with Vera, your AI planning assistant.
Launch Vera
How To Guide
Settings



Voice Input
Push-to-Talk
Audio Stream
Voice Response
OpenAI Realtime API
GPT-4o Realtime
Intent + Tool Call
AI Tool Registry
9 Registered Tools
Code Execution
Application Server
Reactive Processing
API Request
Schedule Data
Posit Connect APIs
P6 Schedule System
UI Refresh

Vera Pilot

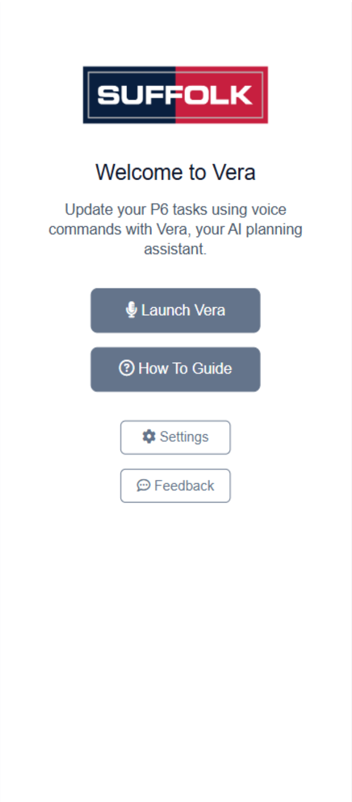
Before AI

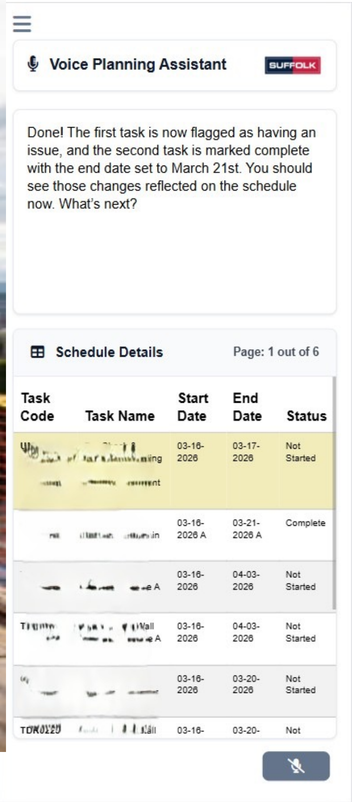
→

→

Vera Pilot

With Vera





SUFFOLK

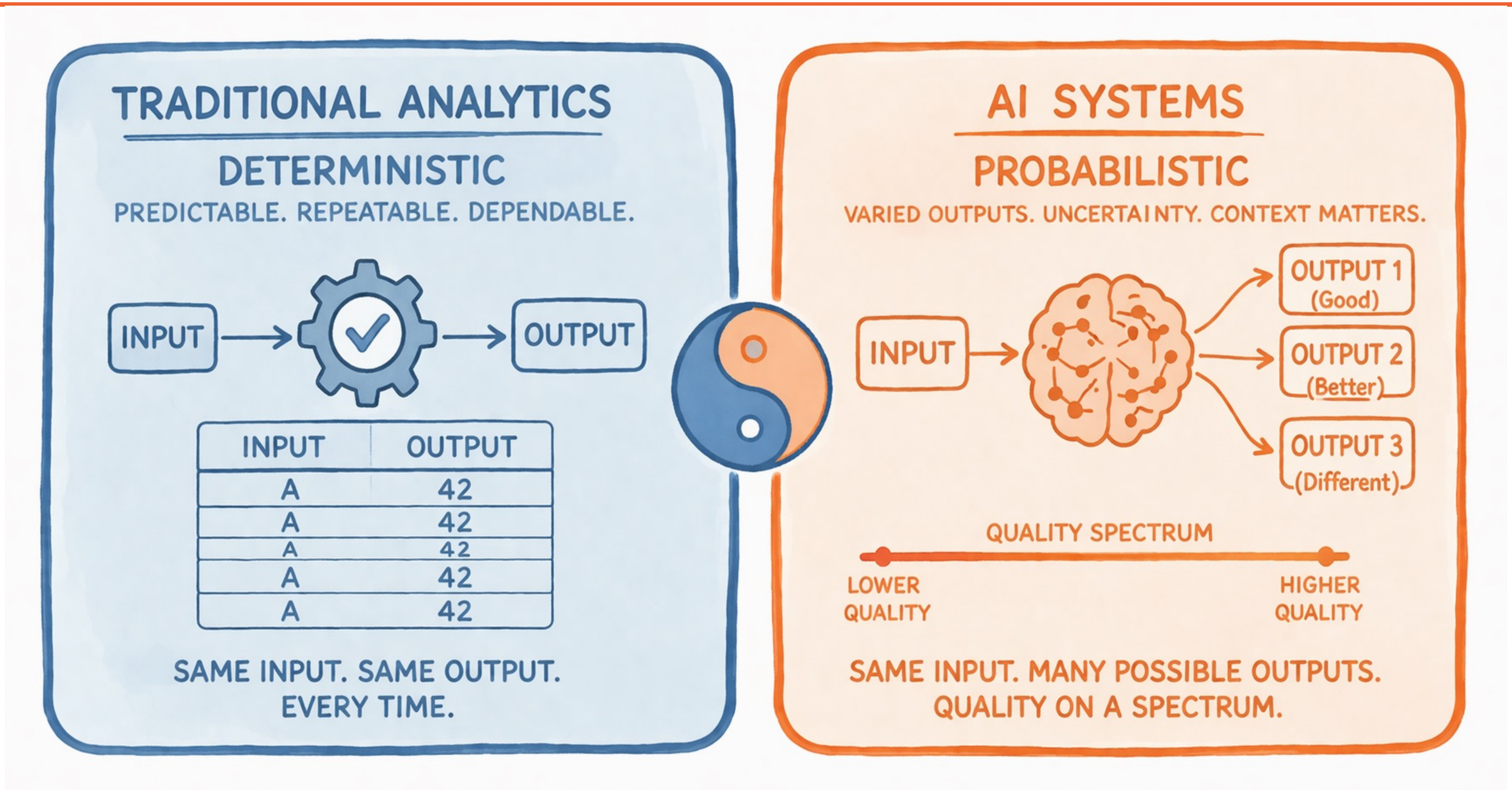
From Streamlining to Improving

The **base case** for LLMs is **streamlining**.

The **future case** for LLMs is **improving**.

How do we get **there** from **here**?

Managing Non-Deterministic Outcomes



This is why **pilot design matters**. You can't just "test and ship."

How the Vera Pilot Managed This

Challenge	Pilot Design Response
Non-deterministic outputs	**Human-in-the-loop** approvals at every step
Overreliance risk	Clear “**draft only**” positioning
Hard to measure quality	**Acceptance-rate tracking** over time
No ground truth	**Side-by-side comparison** to manual process

> *Having a* **baseline comparison** *makes assessment much easier.*

SUFFOLK

From a Technical Perspective

You **can** increase determinism.

Limit Agent Reasoning

- Define explicit **tools**
- Constrain the **action space**
- Provide clear **instructions/guardrails**

Agent as Orchestrator

- Agent as a router, **limit AI creativity**
- **Deterministic pipelines** where possible
- AI reasoning only where it **adds value**

Principle: Reduce the surface area of non-determinism to only where it is needed.

Measuring Success

Benchmarking AI against traditional analytics — not against hype

Let Stakeholders Lead Creatively

Stakeholders bring domain expertise

*But it's up to you **to protect the "how"***

The Scaling Standard

1 **Technical Performance** — Does it work reliably?

2 **Workflow Impact** — Does it improve the process?

3 **Organizational Trust** — Do people believe in it?

Metric
Time-to-decision
Throughput increase
Cost-to-serve
Acceptance rate

Five Principles

1. Start **narrow** and **workflow-specific**
2. Design for **probabilistic systems**
3. **Partner** with stakeholders
4. Measure **operational impact**, not novelty
5. **Earn the right** to expand

Thank You!

Principles for deploying AI that actually works